Distribution of Population

🎯 **In 2006, there was 30 alumina. There is a 10 % increase in every year. What is the distribution. How to find the mean and Standard deviation.**

In the question, the percentage increase is clearly mentioned, which indicates that the population follows an exponential growth model. This is because the quantity increases by a constant percentage (10%) every year. In an exponential growth model, the growth rate remains constant over time.

The general formula is:

$$\boldsymbol{N(t) = N_0(1 + r)^t}$$

Where:

- $N(t)$= number after $t$years
- N_0= initial number
- r= growth rate
- t= number of years

For your question:

- Initial alumni in 2006: $N_0 = 30$
- Growth rate: $r = 10\% = 0.10$

Check the value after 20 years

From 2006 to 2026:

$$\begin{aligned}
&t = 20\\
&N(20) = 30(1.1)^{20}\\
&(1.1)^{20} \approx 6.7275\\
&N(20) \approx 30 \times 6.7275\\
&N(20) \approx 201.8
\end{aligned}$$

So with 10% yearly growth, the alumni count after 20 years would be approximately, 202.

Mean (Average)

Formula:

$$\bar{x} = \frac{\sum x_i}{n} = \frac{Add\ all\ yearly\ Alumina\ counts}{no\ of\ years} = 92.2$$

Standard Deviation

Standard deviation measures how spread out the values are.

Formula:

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

For the 10% growth series, the approximate standard deviation is 52.6. This value indicates that the alumina changes a lot over the year.

Note:

Situation	**Likely Model**
Fixed percentage increase	Geometric / Exponential
Random counts of events	Poisson. Eg: No: of calls arriving per hour
Symmetric random measurements. Values fluctuates unpredictably around central value.	Normal
Success/failure trials	Binomial

🎯 **There is 1.2 million population. What is the sample size for the population to have 95 % Confidence interval.**

Ans: Even though the population size is very large (1.2 million), the sample size does not increase drastically because sample size mainly depends on

1. Confidence level
2. Margin of error
3. Variability in data

There are different sample size formulas depending on the study type.

1. For proportions/percentages

Use**:**

$$\boldsymbol{n = \frac{Z^2 p(1-p)}{e^2}}$$

- $Here, variability\ is\ measured\ using\ p(1-p)$

2. For estimating a mean

Use standard deviation instead**:**

$$\boldsymbol{n = \frac{Z^2 \sigma^2}{e^2}}$$

where**:**

- σ= population standard deviation.
- $Here, variability\ is\ measured\ using\ standard\ deviation.$

To calculate the sample size for a population of **1.2 million** with **95% confidence interval (CI)**, you also need:

- **Margin of error (precision)** — commonly 5%
- **Estimated population proportion (p)** — usually 50% (0.5) if unknown

Using the standard formula for sample size:

$$n = \frac{Z^2 p(1-p)}{e^2}$$

Where:

- $Z = 1.96$for 95% CI
- $p = 0.5$
- $e = 0.05$(5% margin of error)

$$n = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2}$$

$$n \approx 384.16$$

Note:

- **95% confidence level means that if we repeat the study many times, about 95% of the calculated CI would contain the population parameters. CI gives a range within which the true population parameter is likely to fall.**

- **Population parameters: Population mean, Population proportion (fraction or percentage having a characteristic), Population variance σ^2 (spread of population values), Population standard deviation (Avg variation from the mean), population size**

- **Margin of error(e) is the maximum expected difference between the sample estimate and the true population parameter.**

- **Variability in data: Measures how data values differ from each other**

- **From Z table, corresponding to 0.975 (95% + 2.5%) cumulative probability is 1.96**

- **If the margin of error is changed from 5% to 3% OR 1%, the sample size increases, approximately 1067 for 3% and much larger sample size for 1%.**

- **When population size is small, say, N = 500. To find the sample size, first find the initial sample size using any of the above equation of sample size and then use the below method**

$$\boldsymbol{n_{adj} = \frac{n}{1 + \frac{n-1}{N}}}$$